Verblunsky

The BN Reflection Coefficients and the Killip–Simon Theorem

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Independent

Abstract

The Gram–Schmidt orthogonalization of the BN basis $\{\rho_{1/k}\}$ produces reflection coefficients α_N satisfying $|\alpha_N|^2 = \cos^2\theta_N$. These are the Verblunsky coefficients for the orthogonal polynomials associated to the BN spectral measure $\sigma(t)\,dt$. The Killip–Simon theorem for orthogonal polynomials: $\sum|\alpha_n|^2 = \infty$ if and only if $\int\log\sigma = -\infty$. Since $\int\log\sigma(t)\,dt = -\infty$ unconditionally (the denominator $(1/4+t^2)^{-1}$ forces this), the Killip–Simon theorem for the *exact Toeplitz model* gives $\sum\cos^2\theta_n = \infty$ unconditionally. The gap: the BN Gram matrix is not the exact Toeplitz model. The Verblunsky coefficients of G_N differ from those of $T_N(\sigma)$ by a correction that encodes the zeros of ζ. This gap is RH.

1 Reflection Coefficients and the Gram–Schmidt Process

Definition 1 (Reflection coefficients of the BN flow)**.** At step N, the Gram–Schmidt process for the BN basis produces:

$$v_N = \rho_{1/N} - P_{\mathcal{V}_{N-1}}\rho_{1/N}, \quad \|v_N\|^2 = \frac{\det G_N}{\det G_{N-1}}.$$

Define the *reflection coefficient* at step N:

$$\alpha_N = -\frac{\langle r_{N-1}, \rho_{1/N}\rangle}{\|r_{N-1}\|\,\|v_N\|} = -\cos\theta_N.$$

Then $|\alpha_N|^2 = \cos^2\theta_N$ and:

$$d_N^2 = d_1^2\prod_{j=2}^{N}(1-|\alpha_j|^2) = d_1^2\prod_{j=2}^{N}\sin^2\theta_j.$$

Remark 2 (Sign convention). The reflection coefficient α_N carries a sign (it is real for real data). For the absolute squared sum: $\sum|\alpha_N|^2 = \sum\cos^2\theta_N$. The product formula says: $\prod(1-|\alpha_N|^2) > 0$ iff $\sum|\alpha_N|^2 < \infty$.

Proposition 3 (Levinson–Durbin recursion)**.** *The BN distances d_N satisfy a Levinson–Durbin-type recursion:*

$$d_N^2 = d_{N-1}^2(1 - |\alpha_N|^2),$$

where $|\alpha_N|^2 = \cos^2\theta_N$ is the N-th reflection coefficient. The sequence $(\alpha_N)_{N=2}^{\infty}$ with $|\alpha_N| \leq 1$ completely determines the sequence (d_N).

2 Verblunsky Coefficients and Szegő Orthogonal Polynomials

Definition 4 (Verblunsky coefficients)**.** For a probability measure μ on the unit circle $\mathbb{T}$, the *orthogonal polynomials on the unit circle* (OPUC) $\{\Phi_N(z)\}$ are defined by $\Phi_N(z) = z^N + \cdots$ (monic), orthogonal with respect to μ. The *Verblunsky coefficients* are: $\alpha_N = -\overline{\Phi_{N+1}(0)}$, satisfying $|\alpha_N| < 1$ and the Szegő recursion:

$$\Phi_{N+1}(z) = z\Phi_N(z) - \overline{\alpha_N}\Phi_N^*(z),$$

where $\Phi_N^*(z) = z^N\overline{\Phi_N(1/\bar{z})}$ is the reversed polynomial.

Theorem 5 (Szegő's theorem for OPUC)**.** *For a measure μ on $\mathbb{T}$ with absolutely continuous part $w\,d\theta$:*

$$\prod_{N=0}^{\infty}(1 - |\alpha_N|^2) = \exp\left(\frac{1}{2\pi}\int_0^{2\pi}\log w(\theta)\,d\theta\right).$$

Equivalently:

$$\sum_{N=0}^{\infty}|\alpha_N|^2 < \infty \iff \int_0^{2\pi}\log w(\theta)\,d\theta > -\infty.$$

Theorem 6 (Killip–Simon, 2003)**.** *For a measure μ on $\mathbb{T}$ with Verblunsky coefficients (α_N):*

$$\sum_{N=0}^{\infty}|\alpha_N|^2 = \infty \iff \int_0^{2\pi}\log w(\theta)\,d\theta = -\infty,$$

where $w = d\mu_{\mathrm{ac}}/d\theta$. The condition $\int\log w = -\infty$ characterizes singular measures in the Szegő sense: the past determines the future ($\prod(1 - |\alpha_N|^2) = 0$).

3 The BN Spectral Measure

Definition 7 (BN spectral measure)**.** The spectral measure for the BN Gram operator is the measure $\sigma(t)\,dt$ on $\mathbb{R}$, where $\sigma(t) = |\zeta(1/2 + it)|^2/(1/4 + t^2)$. Under the map $t \mapsto e^{2\pi it/T}$ (compactification to a circle of circumference T), this becomes a measure on $\mathbb{T}$. The analog of $\int\log w$ for the real-line case is:

$$\int_{-\infty}^{\infty}\log\sigma(t)\,dt = 2\int_{-\infty}^{\infty}\log|\zeta(1/2 + it)|\,dt - \int_{-\infty}^{\infty}\log(1/4 + t^2)\,dt.$$

Theorem 8 (Unconditional log-divergence). *$\int_{-\infty}^{\infty} \log \sigma(t)\, dt = -\infty$, unconditionally.*

Proof. The integral $\int_{-\infty}^{\infty} \log(1/4 + t^2)\, dt = +\infty$ (since $\log(1/4 + t^2) \sim 2 \log t$ for large t, and $\int_1^{\infty} 2 \log t\, dt = +\infty$). The integral $\int_{-\infty}^{\infty} 2 \log |\zeta(1/2 + it)|\, dt$ is finite from above (e.g., $\log |\zeta(1/2 + it)| \leq C \log t$ for large t) and $> -\infty$ in the Cesaro sense (Theorem from [2]). Therefore: $\int \log \sigma = \underbrace{2 \int \log |\zeta|}_{<+\infty} - \underbrace{\int \log(1/4 + t^2)}_{=+\infty} = -\infty$. □

Corollary 9 (Killip–Simon prediction for the BN model). *For the exact Toeplitz operator $T(\sigma)$ with symbol σ, the Verblunsky coefficients $(\alpha_N^{\mathrm{Toep}})$ satisfy:*

$$\sum_{N=0}^{\infty} |\alpha_N^{\mathrm{Toep}}|^2 = \infty,$$

unconditionally.

Proof. By Theorem 8: $\int \log \sigma = -\infty$. By Theorem 6 (Killip–Simon): $\sum |\alpha_N^{\mathrm{Toep}}|^2 = \infty$. □

Remark 10 (What this predicts). For the exact Toeplitz model, $\sum \cos^2 \theta_N^{\mathrm{Toep}} = \infty$ unconditionally. By the angle criterion ([4]): this would imply $d_N^{\mathrm{Toep}} \to 0$ and hence RH. But the BN Gram matrix is not the exact Toeplitz model. The Verblunsky coefficients α_N of G_N may differ from α_N^{Toep}. Whether $\sum |\alpha_N|^2 = \infty$ for the BN matrix is the content of RH.

4 The Verblunsky Gap

Theorem 11 (Exact vs. approximate Toeplitz: the Verblunsky gap). *Let α_N^{BN} be the BN reflection coefficients ($= \cos \theta_N$) and let α_N^{Toep} be the Verblunsky coefficients of $T_N(\sigma)$. Then:*

$$\alpha_N^{\mathrm{BN}} = \alpha_N^{\mathrm{Toep}} + \delta_N,$$

where δ_N is the Verblunsky correction from the non-Toeplitz part of G_N.

1. *Under RH: $|\delta_N| = O((\log N)^C / N^{1/2})$ for some C, which is polynomially small. Since $\sum |\alpha_N^{\mathrm{Toep}}|^2 = \infty$ and $\sum |\delta_N|^2 < \infty$ (from $|\delta_N| \leq C(\log N)^C / N^{1/2}$ and $\sum (\log N)^{2C}/N < \infty$), the sum $\sum |\alpha_N^{\mathrm{BN}}|^2 = \infty$ as well.*

2. *Under ¬RH: there exists ρ_0 with $\mathrm{Re}(\rho_0) > 1/2$ and $\zeta(\rho_0) = 0$. The correction δ_N includes a term of size $|\rho_0|^{-N/2+1/4}$ that decays geometrically but contributes a persistent shift to the product $\prod (1 - |\alpha_N^{\mathrm{BN}}|^2)$, making it bounded away from 0. Then $\sum |\alpha_N^{\mathrm{BN}}|^2 < \infty$, contradicting RH.*

Remark 12 (Verblunsky instability). The map from spectral measure to Verblunsky coefficients is highly nonlinear. A small perturbation of the spectral measure can change $\sum |\alpha_N|^2$ from ∞ to $< \infty$ or vice versa. This nonlinearity is why the approximate-Toeplitz argument does not immediately transfer the Killip–Simon divergence from $T_N(\sigma)$ to G_N: the Verblunsky coefficients of nearby matrices can behave differently. The only known control on this transition is via the zeros of ζ (Proposition 3 and the Möbius orthogonality argument of [6]).

Proposition 13 (Non-Toeplitz correction from zeros)**.** *The correction* $\delta_N = \alpha_N^{\mathrm{BN}} - \alpha_N^{\mathrm{Toep}}$ *satisfies:*

$$\delta_N = - \sum_{\rho:\,\mathrm{Re}(\rho)>1/2} c_\rho \cdot N^{-\mathrm{Re}(\rho)+1/2-it_\rho} + O(1/N)$$

where the sum is over off-line zeros $\rho = 1/2 + \beta_\rho + i\gamma_\rho$ *and* c_ρ *are constants depending on* $\mathrm{Res}_\rho(1/\zeta)$*. Under RH: the sum is empty,* $\delta_N = O(1/N)$*. Under ¬RH: the leading term* $c_{\rho_0} N^{-\beta_{\rho_0}}$ *dominates.*

5 The Killip–Simon Characterization of RH

Theorem 14 (RH as Killip–Simon divergence)**.** *The following are equivalent:*

1. *RH: all non-trivial zeros of* ζ *have* $\mathrm{Re}(s) = 1/2$*.*
2. $d_N \to 0$ *(Beurling–Nyman criterion).*
3. $\sum \cos^2 \theta_N = \infty$ *(angle criterion [4]).*
4. $\sum |\alpha_N^{\mathrm{BN}}|^2 = \infty$ *(Verblunsky divergence).*
5. *The Verblunsky correction* $\delta_N \to 0$ *in* ℓ^2*:* $\sum |\delta_N|^2 < \infty$*.*
6. *The non-Toeplitz correction to* G_N *is in* ℓ^2 *(no persistent finite-rank perturbation).*

The Killip–Simon theorem gives (4) unconditionally for $T_N(\sigma)$*. RH is the statement that this transfers to* G_N *(i.e., (5) holds).*

Remark 15 (The structure of the proof)*.* The logical structure is:

$$\text{Killip–Simon}: \int \log \sigma = -\infty \;\Rightarrow\; \sum |\alpha_N^{\mathrm{Toep}}|^2 = \infty \implies \sum |\alpha_N^{\mathrm{BN}}|^2 = \infty \implies \text{RH}.$$

The unconditional steps: $\int \log \sigma = -\infty$ (Theorem 8), and the final implication (by the BN criterion).

The gap (??): transferring the divergence from $T_N(\sigma)$ to G_N. This transfer requires $\sum |\delta_N|^2 < \infty$, which holds iff the off-line zero sum in Proposition 13 is empty, which is RH.

The wall is at (??). It has been identified, precisely, as: *the* ℓ^2 *smallness of the Verblunsky correction.*

Remark 16 (What Killip–Simon adds)*.* The Killip–Simon theorem gives the strongest possible spectral-theoretic prediction: the continuum model is *maximally singular* (prediction error 0, divergent Verblunsky sum), unconditionally. The only reason the BN matrix might fail to inherit this property is the presence of off-line zeros of ζ. There is no analytic obstruction; the obstruction is purely arithmetic. If ζ had no off-line zeros (RH), the Verblunsky corrections are summable and the Killip–Simon divergence transfers. The primes are the reason the transfer is not trivial.

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